

Financial Contracts

My friend Jerry lives in a small town, the same town in which he was born and raised. Because Jerry's parents are no longer alive and many of his friends have left, he is seriously thinking of packing up and moving to a larger city. However, Jerry recently heard that there is a plan to build a major highway that will pass very close to his hometown. Because the highway is likely to bring new life to the town, Jerry is reconsidering his decision to move away. It has also occurred to Jerry that the highway may bring new business opportunities.

For many years, Jerry's family was in the restaurant business, and Jerry is thinking of building a restaurant at the main intersection leading from the highway into town. If Jerry does decide to build the restaurant, he will need to acquire land along the highway. Fortunately, Jerry has located a plot of land, currently owned by Farmer Smith, that is ideally suited for the restaurant. Because the land does not seem to be in use, Jerry is hoping that Farmer Smith might be willing to sell it.

If Farmer Smith is indeed willing to sell, how can Jerry acquire the land on which to build his restaurant? First, Jerry must find out how much Farmer Smith wants for the land. Let's say \$100,000. If Jerry thinks that the price is reasonable, he can agree to pay this amount and, in return, take ownership of the land. In this case, Jerry and Farmer Smith will have entered into a *spot* or *cash transaction*.

In a cash transaction, both parties agree on terms, followed immediately by an exchange of money for goods. The trading of stock on an exchange is usually considered to be a cash transaction: the buyer and seller agree on the price, the buyer pays the seller, and the seller delivers the stock. The actions essentially take place simultaneously. (Admittedly, on most stock exchanges, there is a settlement period between the time the price is agreed on and the time the stock is actually delivered and payment is made. However, the settlement period is relatively short, so for practical purposes most traders consider this a cash transaction.)

However, it has also occurred to Jerry that it will probably take several years to build the highway. Because Jerry wants the opening of his restaurant

Forward Pricing

What should be the fair price for a forward contract? We can answer this question by considering the costs and benefits of buying now compared with buying on some future date. In a forward contract, the costs and benefits are not eliminated; they are simply deferred. They should therefore be reflected in the forward price.

$$\text{forward price} = \text{current cash price} + \text{costs of buying now} \\ - \text{benefits of buying now}$$

Let's return to our example from Chapter 1 where my friend Jerry wanted to acquire land on which to build a restaurant. He was considering both a cash purchase and a one-year forward contract. If he enters into a forward contract, what should be a fair one-year forward price for the land?

If Jerry wants to buy the land right now, he will have to pay Farmer Smith's asking price of \$100,000. However, in researching the feasibility of a one-year forward contract, Jerry has learned the following:

1. The cost of money, whether borrowing or lending,¹ is currently 8.00 percent annually.
2. The owner of the land must pay \$2,000 in real estate taxes; the taxes are due in nine months.
3. There is a small oil well on the land that pumps oil at the rate of \$500 per month; the oil revenue is receivable at the end of each month.

If Jerry decides to buy the land now, what are the costs compared with buying the land one year from now? First, Jerry will have to borrow \$100,000 from the local bank. At a rate of 8 percent, the one-year interest costs will be

$$8\% \times \$100,000 = \$8,000$$

¹ At this point, we will assume that the same interest rate applies to all transactions, whether borrowing or lending. Admittedly, for a trader, the interest cost of borrowing will almost always be higher than the interest earned when lending.

Figure 7-2 Effect of changing interest rates on option values.

	If domestic rates rise	If domestic rates fall	If foreign rates rise	If foreign rates fall
stock option calls will	rise	fall	not applicable	not applicable
stock option puts will	fall	rise	not applicable	not applicable
futures option calls (stock-type settlement)	fall	fall	not applicable	not applicable
futures option puts (stock-type settlement)	fall	fall	not applicable	not applicable
futures option calls (futures-type settlement)	no	no	not applicable	not applicable
futures option puts (futures-type settlement)	effect	effect	not applicable	not applicable
foreign currency option calls	rise	fall	fall	rise
foreign currency option puts	fall	rise	rise	fall

Figure 7-3 Effect of changing dividends on stock option values.

	If the dividend is raised	If the dividend is cut
stock option calls will	fall	rise
stock option puts will	rise	fall

the *partial derivatives*, will not answer all our questions concerning changing market conditions, but they are an important starting point in analyzing the risks associated with both simple and complex option positions.

The Delta

The *delta* (Δ) is a measure of an option's risk with respect to the direction of movement in the underlying contract. A positive delta indicates a desire for upward movement; a negative delta indicates a desire for downward movement. The delta has several different interpretations, any of which may be useful to a trader depending on the types of strategies being executed.

Rate of Change

At expiration, an option is worth exactly its intrinsic value. Prior to expiration, however, the theoretical value of an option is a curve that will approach intrinsic value as the option goes very deeply into the money or very far out of the

Using (4.18) and (4.20), respectively, the *likelihood of the D-vine copula* is given by

$$\begin{aligned} \ell(\theta, \mathbf{u}) = & \prod_{k=1}^n \prod_{j=1}^{d-1} \prod_{i=1}^{d-j} \\ & \times c_{i,i+j;i+1:i+j-1}(C_{i|i+1:i+j-1}(u_{k,i}|u_{k,i+1:i+j-1}), C_{i+j|i+1:i+j-1}(u_{k,i+j}| \\ & u_{k,i+1:i+j-1})), \end{aligned} \quad (7.3)$$

while the *likelihood of a C-vine copula* can be written as

$$\begin{aligned} \ell(\theta, \mathbf{u}) = & \prod_{k=1}^n \prod_{j=1}^{d-1} \prod_{i=1}^{d-j} \\ & \times c_{j,i+j;1:j-1}(C_{i|1:j-1}(u_{k,i}|u_{k,1:j-1}), C_{i+j|1:j-1}(u_{k,i+j}|u_{k,1:j-1})). \end{aligned} \quad (7.4)$$

In particular for D-vine copulas, the conditional distribution function $C_{i|i+1:i+j-1}$ depends first on $\theta_{i,i+j-1;i+j-2}$, while $C_{i+j|i+1:i+j-1}$ on $\theta_{i+2,i+j-1;i+j-1}$. This can be used recursively to determine the complete parameter dependence.

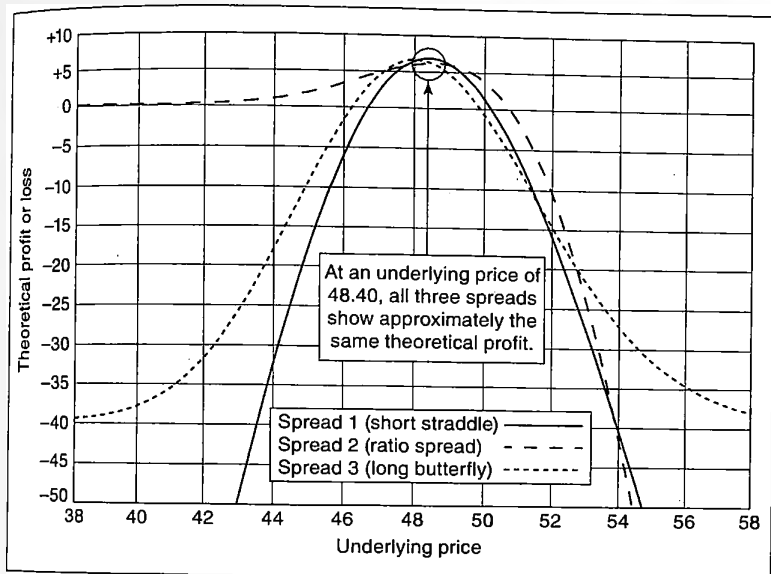
7.2 Sequential and Maximum Likelihood Estimation in Simplified Regular Vine Copulas

The standard parameter estimation method in statistics is maximum likelihood. This involves maximizing the likelihood. Often, it is easier to maximize the log-likelihood instead of the likelihood. Here it is particularly easy, since the likelihood of a regular vine copula is a product of terms, which transforms to a sum, when we optimize the log-likelihood. For hierarchical Archimedean copulas (see for example Savu and Tiede 2010 and Okhrin et al. 2013), this is not possible.

In the case of simplified regular vine copulas in d dimensions with pair copulas with a single parameter, the number of parameters to be maximized is $d(d-1)/2$, which grows quadratically. Therefore it is important to find good starting values for the optimization. Such starting values are given by sequential parameter estimates. We first illustrate this approach in the three dimensional case.

Ex 7.3 (Sequential estimation in three dimensions) The associated likelihood for the three-dimensional case is given in (7.2). In the first step, we find parameter estimates of θ_{12} and θ_{23} by using the bivariate sub samples $(u_{k,1}, u_{k,2})$ and

Figure 13-4

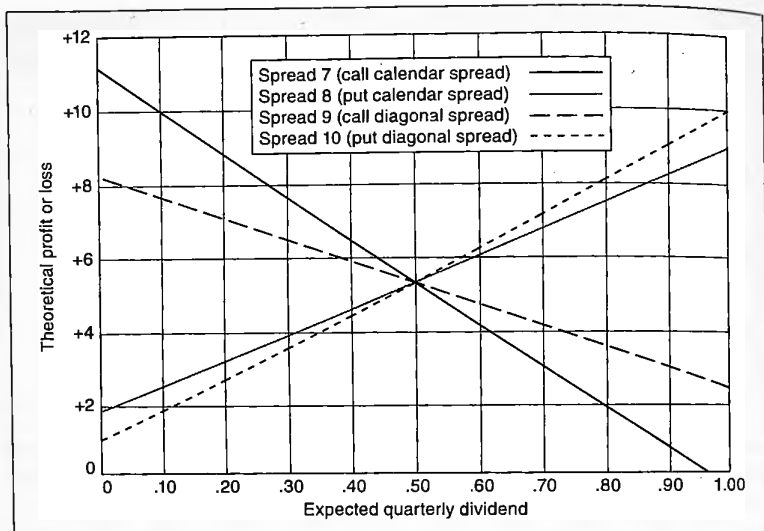


Which spread is best? That depends on what the trader is worried about. If the trader is oblivious to the risk, it won't matter which spread he chooses. On average, each position will show a profit of approximately 6.00. If, however, the trader is more worried about a large downward move in the market, then perhaps Spread 2 is best. And if the trader is unwilling to accept the risk of unlimited loss in either direction, then perhaps Spread 3 is best.

In addition to the possibility of a large move, all three positions are exposed to the risk of an incorrect volatility estimate. Because each spread has a negative vega, there will be no problem if, over the life of the option, volatility turns out to be lower than 18 percent. In such a case, the spreads will show a profit greater than originally expected. On the other hand, if volatility turns out to be greater than 18 percent, this could present a problem. What will happen if volatility turns out to be 20 or 25 percent or some higher number? Each spread will be hurt because of the negative vega, but will they be hurt to the same degree?

Because Spread 2 has the smallest vega (-0.875), we might initially conclude that it has the smallest volatility risk. But the vega, like the gamma, changes as market conditions change. If we raise volatility, the vega of Spread 1, the short straddle, will remain essentially unchanged because the vega of an at-the-money option is constant with respect to changes in volatility. But the vega of Spread 3, the long butterfly, will decline because the vega of in-the-money and out-of-the-money options (the May 46 and May 50 puts) will tend to increase as volatility rises. With Spread 2, the vega of both options, the May 50 call and the May 52 call, will begin to increase, so it's not immediately clear what will happen if we increase volatility.

Figure 13-16 Dividend sensitivity.



gamma, the potential profit when the underlying market moves. The risk is the theta, the money that will be lost through the passage of time if the underlying market fails to make sufficiently large moves. The trader would like the reward (the gamma) to be as large as possible compared with the risk (the theta). We might express this relationship as a ratio

$$\text{gamma}/\text{theta}$$

The larger the absolute value of this ratio, the more efficient the position.

In the same way, a trader who has a negative gamma and a positive theta wants the risk (the gamma) to be as small as possible compared with the reward (the theta). He therefore wants the absolute value of the gamma/theta ratio to be as large as possible.

For example, we might go back and calculate the efficiency of Spreads 1 through 3 in Figure 13-3. The efficiencies are

	Gamma/Theta	Efficiency
Spread 1	-406.0/0.4235	959
Spread 2	-165.5/0.1365	1,212
Spread 3	-370.0/0.4000	925

Because each spread has a negative gamma and positive theta, we want the efficiency to be as small as possible. We can see that Spread 3 is best, which is consistent with our previous analysis of each spread.

Assuming that all strategies have approximately the same theoretical edge, the efficiency can be a reasonable method of quickly comparing strategies where all options expire at the same time. In such cases, the gamma and theta

will cause the option's value to rise by $0.5 \times 0.10 = 0.05$. Consequently, the option will only decline by 0.15, a decline of 0.20 due to a change in the underlying price combined with an increase of 0.05 due to the increase in implied volatility. The option has a *skewed* or *adjusted delta* of -15 .

The inclusion of a volatility skew in a pricing model will affect the calculation of all option risk measures and can greatly complicate a trader's ability to manage risk. For many traders, it may be best to keep things simple, perhaps using a skew model to generate theoretical values while using a traditional model to calculate the delta, gamma, theta, and vega. For an active trader who carries large option positions, calculating accurate skewed sensitivities becomes much more important because the total value of the position can change very quickly as market conditions change. Financial engineers at professional option trading firms are often responsible for developing methods to accurately calculate theoretical values and risk sensitivities using a volatility skew model. But even the most sophisticated model is unlikely to generate values that exactly model option prices under all market conditions. A model can help, but it will always have limitations.

By combining the volatility term structure across expiration dates with the volatility skew across exercise prices, we can form a *volatility surface*. While sometimes difficult to visualize, a volatility surface may enable a trader to more easily see the basic volatility characteristics of an option market. The more exercise prices and expiration dates that are available, the more accurate will be the volatility surface. Sample volatility surfaces for the FTSE 100 options and wheat options are shown in Figures 24-12 and 24-13. At the time, the FTSE 100 Index was trading at 5,966, and the front-month wheat futures contract was trading at 647.

Figure 24-12 FTSE 100 volatility surface, March 16, 2012.

